

MACHINE LEARNING

OPEN ENDED LAB

CAR PRICE PRIDITION

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Scope

This report outlines the development of a Car Price Prediction system using machine learning (ML) techniques. The project encompassed comprehensive stages including data collection, preprocessing, feature extraction, model training, evaluation, and deployment. The system was designed to estimate car prices based on diverse input features, providing users with a tool to assess vehicle costs effectively.

Objective

The primary objective of this project is to develop a machine learning model that accurately predicts the fair market value of used cars based on a set of relevant features.

Development Tools

- **Programming Language:** Python
- **Development Environment:** Google Colab, VS Code
- **Libraries:**
 - **Data Processing:** numpy, pandas
 - **Visualization:** matplotlib, seaborn
 - **Machine Learning:** scikit-learn
 - **Web Interface:** streamlit, streamlit-option-menu
- **Data Source:** Quikr

Development Steps

1. Data Collection

Dataset was scraped from Quikr, specifically car prices of used cars. These datasets provided comprehensive features necessary for training robust predictive models.

2. Data Preprocessing

Data preprocessing was critical to ensure the quality and usability of the data. Steps included:

- **Cleaning:** Removing missing or/and incorrect values
- **Encoding:** Converting categorical variables into numerical values suitable for model training with One Hot Encoder, and Label Encoder libraries.

3. Exploratory Data Analysis (EDA):

EDA was performed to uncover insights from the data. Visualizations such as histograms, box plots, and scatter plots were used to explore the distribution of features and their relationships with each other and with the output (Price). This step helped in understanding the underlying patterns and informed feature selection and engineering.

4. Feature Engineering:

Scaling was done on *kms_driven* and *year* columns using MinMax scalar before training the Linear Regressor. Scaling is often done to normalize the range of features, particularly when using models that are sensitive to the scale of the data. Linear Regression, for instance, isn't as sensitive to the scale of features as some other models like Support Vector Machines (SVM) or K-Nearest Neighbors (KNN). This means that, theoretically, Linear Regression can handle features with different scales without a significant impact on its performance.

However, Scaling can improve the numerical stability of the optimization process used to find the best-fitting line. This is because the optimization algorithms (such as gradient descent) often perform better when the input data is on a similar scale. It can also lead to faster convergence of the algorithm as well as it can make model coefficients more interpretable.

5. Model Building:

Threemachine learning algorithms were chosen for model building:

- **Linear Regression:** Effective for binary classification with linear decision boundaries and regularization to prevent overfitting.
- **Decision Tree:** Suitable for high-dimensional spaces and capable of handling non-linear boundaries using kernel functions.
- **Random Forest:** Combines multiple decision trees to create an ensemble model, improving prediction accuracy and handling non-linear relationships between features and price.

6. Model Training:

The models were trained using the preprocessed datasets.

- **Linear Regression**

The model was trained by first encoding the feature variables (One hot encoder for the model implemented with package and Label Encoder for the one implemented without package). The train test split was also randomized to achieve the greatest R^2 score.

- **Decision Tree Regressor**

Used cross-validation to ensure the robustness of the model using grid search, which tuned the hyperparameters to optimize model performance.

- **Random Forest**

Grid search was also used here to optimize the parameters and train the model.

7. Model Evaluation:

Models were evaluated based on their R^2 score, mean squared error, root mean squared error and mean absolute error. Linear Regression, Decision Tree and Random Forest were compared to determine the best-performing model for each price prediction task.

Model	R^2 Score (%)
Linear Regression(with python package)	91.65
Linear Regression(without python package)	57.88
Random Forest(with python package)	80.71
Random Forest(without python package)	50.59
Decision Tree (with python package)	38.69
Decision Tree (without python package)	31.62

8. Integration:

The linear regressor was chosen as the best model for car price prediction and was integrated into a user-friendly Streamlit application. This web-based interface allows users to easily input the car attributes and get an estimate for the price.

- **Input Forms:** Tailored input forms guide users to enter relevant car data within specified ranges, making the process straightforward and user-friendly.
- **Real-Time Predictions:** Users receive instant feedback on their of price after submitting their data.

Conclusion

The Car Price Prediction System effectively utilized machine learning algorithms to estimate car prices based on various features. Among the models evaluated, the Linear Regression model demonstrated the highest accuracy. Despite this, all models exhibited substantial error metrics (Mean Squared Error, Root Mean Squared Error, and Mean Absolute Error). This is primarily due to the fact that car prices are in the range of hundreds of thousands, meaning even minor deviations can lead to significantly high error values. Consequently, while the Linear Regression model provided the best performance, the inherent scale of car prices amplified the error metrics across all models.

Future Advancements

- **Incorporate Domain Knowledge:** Utilize domain-specific knowledge to improve feature selection and engineering.
- **Outlier Detection and Removal:** Identify and remove or adjust outliers in the dataset. Outliers can skew the model and increase error metrics.
- **Ensemble Methods:** Use stacking to combine multiple models to form a stronger predictive model.
- **Regularization:** Apply regularization techniques (L1 or L2 regularization) to prevent overfitting, which can improve the model's generalization performance and reduce errors.
- **Enhance User Interface:** Improve the interface to make it more intuitive and accessible, possibly adding more interactive features and visualizations.

Snapshot Of Interface

Car Price Prediction

Car Model

Ford Figo

Brand

Ford

Year

2014

-

+

Kilometers Driven

150000

-

+

Fuel Type

Diesel

▼

Predict

The predicted price of the car is: \$183,702.50